1 MOC Development – "Cookbook"

1.1 Overview

You use the "Cookbook" for the development on the MOC. It explains many use cases step by step. Some of the use cases occur frequently, others rarely. The requirements and all steps of the solution are listed in short form. Frequent errors are mentioned so that you can easily avoid these errors. Also note the further information specified in each case.

1.2 Enable the MES Development Suite

Enable the MES Development Suite to make changes on the MOC.

Requirements

- MDS license
- Function authorization "mds"
- Select the correct scope

Solution: Steps to follow

- Go to the main menu "Extras" - "MES Development Suite" and enable the MES Development Suite.
- Reopen the application you want to configure.
- Make changes and save the changes made.

Restrictions

The available MDS license specifies the scope of changes you can make.

1.3 Formatting of durations as axis values in the chart

In the chart, the axis labeling shows numeric values. Actually, the values are durations.

Requirements

- Application including configured chart.

Solution: Steps to follow

1. Open the configuration file of the chart in a text editor.
2. Edit the value for "CustomFunction" and change it to "Duration".

```
<Setting Key="CustomFunction" Description="" LastChanged="2015-07-15T12:51:32.8967665Z" ValueType="System.String"
Version="0.0.0.0">
```

```
    <Value>
      <string>Duration</string>
    </Value>
</Setting>
```

3. Save the configuration file.
4. Restart the MOC.

Restrictions

The described solution only works with values that you can format with the output format `mpdv_timespan` e.g. in the grid.

Further information

The respective configuration file is stored in the application directory and uses the naming convention [PluginId]Chart.config.

1.4 Configuration of selection fields

You can store search applications or selection lists for a field in order to simplify the selection of entries. Depending on the use, three options are available for search applications and selection lists:

- Use **search applications** to select a data record from a large quantity of data, e.g. all orders. Search applications include filter criteria that help to narrow down the data.
- For small quantities of data, e.g. all order statuses, define a **dynamic selection list** including the request of a service. The static selection lists do not include filter criteria. These lists always show all data for selection.
- Define a **static selection list** with a ReferenceData if you only want to offer few specified values.

Requirements

- "MDS Development Suite" is enabled to configure applications.
- You have selected the correct scope to configure applications.
- You have created the field you want to configure in the detail application (see tutorial: "Adding and customizing fields in a detail application").
- The data you want to store (search application, data of selection list) must be available in the server or client.

Solution: Steps to follow

Configuration of a search application

Store a button for the field to open an application and select an entry from an existing application, e.g. open the order overview to transfer an order into the field.

1. Open the application you want to modify.

2. Right-click the layout of the detail application - "Customize layout"
3. Select the field you want to configure.
4. "ControlDataSource": Assign the application to be opened, e.g. `OrderOverview`.
5. "ControlDataSourceMode": Set `Lookup`
6. Select the parameters "ControlDataSourceParameter" that are transferred to the application to be opened, e.g. a specified order number via `order.id=12002500`
7. Set the "ControlDataSourceResult", e.g. `order.id;order.id;;order.articledesignation;order.act.status`
   - the internal (unique) ID, here `order.id`
   - the value to be transferred into the field, here `order.id`
   - the text to be placed behind the field, usually units like e.g. `n hours`, here empty
   - further parameters that can be displayed in other fields of the application, here `order.articledesignation;order.act.status`
8. [Optional] "ControlParameter": Assign the DataLogic of the service that is to be called when the user directly types into this field and the field of the service where the entries are made, here e.g. `BOOrderList;order.id`
9. "ControlType": Assign `TextEdit`
10. Save the application by clicking the button "Save".

Note: If you want to use the field including search application at different places on the MOC, it might be useful to make the described configuration in the higher-level repository client, and not in the client.

Configuration of a dynamic selection list with data of a service

Store a selection list for a field to select a value from a limited number of database entries, e.g. all order statuses. The entries in the list are requested dynamically from the service.

1. Define the ControlDataSource in the repository.
   1. Create a new data record in the view "ControlDataSource".
   2. Select the domain and service (as DataLogic in the field Source), which provide the data, e.g. `MDOrderStatus` and `MDOrderStatsList`
   3. Define the "name" of the ControlDataSource. Then this name will be used in the client, e.g. `orderStatusList`
   4. Define the "columns" to request, e.g. `orderstatus.status;orderstatus.text`
   5. Define the "parameters" that limit the results, e.g. `orderstatus.statustype=A`
   6. Specify the "Result"
      - the internal (unique) ID, here `order.id`
      - the value to be transferred into the field, here `order.id`
      - the text to be placed behind the field, usually units like e.g. `n hours`, here empty
      - further parameters that can be used in the application, here `order.articledesignation;order.act.status`

7. Test deployment of the ControlDataSource (see tutorial "Deployment of customizations")

2. Open the application that you want to modify on the MOC.
3. Right-click the layout of the detail application - "Customize layout"
4. Select the field you want to configure.
    1. "ControlDataSource": Assign the name of the ControlDataSource defined in the repository.
    2. "ControlDataSourceMode": Set `Lookup`
    3. Set the "ControlDataSourceResult", if it is not identical to the data defined in the repository (syntax as described above).
    4. "ControlType": Set `ComboBoxEdit`
    5. "ControlTypeMode": Set e.g. `SingleEdit` for single selection or `Multiple` for multiple selection.
5. Save the application by clicking the button "Save".

Configuration of a static selection list with predefined selection

Store a predefined selection list in a field as ReferenceData, e.g. order types. Unlike the ControlDataSource, the ReferenceData is a predefined list of values that the user cannot change.

1. Define the ReferenceData in the repository.
    1. Create a new data record for each required selection option in the view "ReferenceData".
    2. Select the domain providing the data, e.g. `MDOrderType`
    3. Define the "type" of the ReferenceData. This type will be used as ReferenceData name in the client, e.g. `checksendaheadtype`.
    4. Define the "db_key" as the actual value of the concrete data record, e.g. `N`.
    5. Define the "ref_data_key" as the combined value of "type" and "db_key", e.g. `checksendaheadtype:N`
    6. Define the default value ("is_default"), the designation ("designation") and the sort sequence ("sort_key")
    7. Test deployment of the ReferenceData (see tutorial "Deployment of customizations")
2. Open the application that you want to modify on the MOC.
3. Right-click the layout of the detail application - "Customize layout"
4. Select the field you want to configure.
    1. "ControlDataSource": Assign the name of the ReferenceData defined in the repository.
    2. "ControlDataSourceMode": Set `Reference`
    3. Set the "ControlDataSourceResult", if it is not identical to the data defined in the repository (syntax as described above).
    4. "ControlType": Set `ComboBoxEdit` or `RadioGroup`
    5. "ControlTypeMode": Set e.g. `Single` for single selection or `Multiple` for multiple selection.

5. Save the application by clicking the button "Save".

Affected files

The changes described here affect the following files:

- In the relevant application directory `<MOC>\local\conf\MOC\Apps\`, the file `Layout<ID of the detail application>.config` of the respective detail application.
- In the MOC main directory, the file `<MOC>\custom\resources\data\controldatasources\<ID of the application>.ControlDataSources.xml` when defining ControlDataSources
- There is this file on the server `<InstallDir>\jdir\MOC\1\referenceData\local\<Service-Domain>.xml` if you configure ReferenceData

Possible problems

- In case of configuration errors, you can usually find the reason in the log files, e.g. typing errors in service names, etc.

Further information

- MDS-BAS_81: Customizing of main applications, section "Selection panel"
- MDS-BAS_81: Input and output fields (to assign field properties)
- MDS-BAS_81: Sections "ControlDataSource" and "ReferenceData"
- Tutorial: "Adding and customizing fields in a detail application"
- Tutorial: "Deployment of customizations"
- Tutorial: "Adding and customizing fields in a detail application"

1.5 Deployment of customizations

Further information

- CUT-MOC: Documentation of the training CUT-MOC: Deployment for artifacts of client and server.
- MDS-BAS_81: MOC Update Package Creator
- MDS-BAS_81: Update Packages for the Maintenance Manager

1.6 Generate a new application

Create a new application or create an application including editing applications (insert, update, delete, copy)

Requirements

- "MDS Development Suite" is enabled to configure applications.
- You have selected the correct scope to configure applications.

Solution: Steps to follow

Create new main application

1. Go to: MOC "MES Development Suite" - "New application"
2. Save the new application by clicking the button "Save".
   - Define a unique ID for the application, e.g. `U_myOrders`
   - Define a caption including language key, e.g. `lkU_myorders`
   - (Optional) Specify a help file and index, define the update rate to reload data and application icon.
3. Customize the application and define selection panel, data sources, detail applications, toolbar (see the corresponding tutorials).
4. Save the application by clicking the button "Save".

Creating an application with editing dialogs

1. Go to: MOC "MES Development Suite" - "Generate application"
2. Select the underlying DataLogic (usually of type list or overview)
3. Check the predefined fields for the editing dialogs you want to create
4. If necessary, define additional dialogs in the tab "Additional functions"
5. Generate the application by clicking "Generate".
6. Define the application properties.
   - Define a unique ID for the application, e.g. `U_myOrders`
   - Define a caption including language key, e.g. `lkU_myorders`
   - (Optional) Specify a help file and index, define the update rate to reload data and application icon.
7. Save the application by clicking the button "Save".

Affected files

If you generate an application, a new directory with the ID of the new application is created in the directory `<MOC>\local\conf\MOC\Apps\`. All files an application requires are stored here.

If you generate an application including editing dialogs, the application directory will include an additional directory for each editing dialog. All configuration files required for the respective editing application are stored in these directories.

Possible problems

- New applications are not automatically integrated into the menu. By default, you can only call the application using the transaction code `_capp id=<ID of the application>`. The tutorial "Create or edit the menu" describes how to integrate an application into the menu.
- You cannot add editing dialogs subsequently to an application. This is only possible using the application generator. You can directly change the configuration files if you only want to make minor changes to the editing applications, . After having changed the files, you must reload the configuration data (MOC menu "MES Development Suite" - "Reload configuration data").

Further information

- MDS-BAS_81: Customizing of main applications, section "Detail applications"
- MDS-BAS_81: Customizing of editing applications

1.7 Defining a conditional formatting in the grid

Configure a conditional presentation in the grid to simplify the display of information.

Requirements

- "MDS Development Suite" is enabled to configure applications.
- You have selected the correct scope.
- The detail application of type "grid" is configured.

Solution: Steps to follow

1. Open the application.
2. Right-click the grid column - "Configure conditional display".
3. "Add" a new configuration.
    - Configure the required display via "Appearance", e.g. red as "BackColor".
    - Configure the condition via "AppearanceDescription":
        - "Column": The condition is valid for the selected columns.
        - "Condition": Defines the type of condition to be checked, e.g. `Between`
        - "Value1" or "Value2" define the limit values, e.g. `100` as lower, `300` as upper limit.
4. Save the changed application by clicking the button "Save".

Refer to the documentation for detailed information on how to configure the conditions to be checked.

Affected files

The changes described here affect the following files in the respective application directory `<MOC>\local\conf\MOC\Apps\`:

- The file `Grid<ID of the grid detail application>.config` as layout file of the detail application of type "grid".

Further information

- MDS-BAS_81: Components for detail applications, section "Table view (grid)"

1.8 Configuration of column totals in the grid

Configure totals per grid column or per grid column and group to simplify the display of information.

Requirements

- "MDS Development Suite" is enabled to configure applications.
- You have selected the correct scope.
- The detail application of type "grid" is configured.

Solution: Steps to follow

1. Open the application.
2. Right-click the grid column - "Grid properties"
3. Configure the property "SummaryItem"
   - SummaryType: Sum (most common), max, min, etc.
   - Special case: Use the following configuration with durations: Display format: {0:mpdv_timespan}; SummaryType: Custom; Tag: TimeSpanSum
4. Right-click the grid column - "Show overall column totals" to obtain a total of all data.
5. Or: Right-click the grid column - "Show column totals for group" to obtain a total per group.
6. Save the changed application by clicking the button "Save".
7. Reopen the application and request data.

Affected files

The changes described here affect the following files in the respective application directory `<MOC>\local\conf\MOC\Apps\`:

- The file `Grid<ID of the grid detail application>.config` as layout file of the detail application of type "grid".

Further information

- MDS-BAS_81: Components for detail applications, section "Table view (grid)"

1.9 Grouping fields within a detail application

Group individual fields to improve the presentation within a detail application of type "layout".

Requirements

- "MDS Development Suite" is enabled to configure applications.
- You have selected the correct scope to configure applications.
- The fields you want to group are defined in the detail application of type "layout".

Solution: Steps to follow

Group fields in groups or tabs.

1. Open the application you want to modify.
2. Right-click the layout of the detail application - "Customize layout"
3. In the dialog "Customization", change to "Layout tree view".
4. In the tree view, select all fields that you want to group (e.g. via CTRL + click).
5. In the context menu, select "Group" or "Create tabbed group".
6. Define a language key for the newly created group in the field `CustomizationFormText` of the respective control.
7. Save the application by clicking the button "Save".

Remove grouping of fields

1. Open the application you want to modify.
2. Right-click the layout of the detail application - "Customize layout"
3. In the dialog "Customization", change to "Layout tree view".
4. Select the group or the tab you want to remove.
5. In the context menu, select "Clear grouping" or "Clear tabbed group".
6. Save the application by clicking the button "Save".

Affected files

The changes described here affect the following files in the respective application directory `<MOC>\local\conf\MOC\Apps\`:

- The file `Layout<ID of the detail application>.config` if the detail application has been customized.

Further information

- MDS-BAS_81: Customizing of main applications, section "Selection panel"
- Tutorial: "Adding and customizing fields in a detail application"

1.10 Creating and modifying label texts

You can create or modify label texts for MOC applications to integrate customer-specific terminology or translations.

Requirements

- "MES Development Suite" is enabled.
- The MPDV custom dictionary `mpdvDictionaryCustomer.xlm` is available.

Solution: Steps to follow

Customize an existing language key.

1. Identify the language key you want to edit in the respective MOC application.
   - The key refers to the name of the application:
     - Open the dialog "Configure application" by clicking the button in the toolbar.
     - The field value of "caption" is e.g. `lkOrderOverview`
   - The key refers to the name or the tooltip of a button:
     - Open the "Link editor" by calling the context menu in the toolbar and clicking "Configuration".
     - The field value of "Title" or "Tooltip" is e.g. `lkInterrupt`
   - The key refers to a field label in the selection panel or in the detail application of type "layout":
     - Open the dialog "Customization" by right-clicking the field label and selecting "Customize layout".
     - Click and select the field while "Customization" dialog is open.
     - The field value of "LanguageKey" is e.g. `lkOrder`
2. Enter the new text for the identified language key in the custom dictionary in form of a new entry.
3. Create the language resource files by clicking "Create resource" in the Excel file and store the files in `<MOC directory>\local\resources\languages`
4. Restart the MOC.

Define a new language key

1. Create a new entry in the custom dictionary and assign a unique key, e.g. `lkU_inspection`
2. Check if a duplicate of the key exists by clicking "Check dup. keys" in the Excel file.
3. Create the language resource files by clicking "Create resource" in the Excel file and store the files in `<MOC directory>\local\resources\languages`. Refer to the documentation for further details on the languages you can select.
4. Enter the new language key in the MOC application.
5. Restart the MOC.

Affected files

The changes described here affect the following files:
- Language files in `<MOC directory>\local\resources\languages`

Restrictions

- You cannot assign language keys to user fields in the individual applications. This is only possible in the user field configuration.
- You must change the language keys of column headers of a grid in the grid configuration. Right-click the grid column and select "Grid properties". Change the property "LanguageKey" of the respective field in the configuration. If you want to change the language key for all identical fields, we recommend to make the configuration in the repository.

Possible problems

- Problem: The newly configured language key is not translated in the application, but `lkU_xxx` is displayed.
    - Proceed as follows: Make sure that the language key is spelled properly (case sensitive) and that the language files are exported to the correct directory. Then restart MOC.
- Problem: The language key is translated, but is displayed in the wrong language.
    - Proceed as follows: Make sure that the language selected in the MOC has been exported and that the file is stored in the correct directory.
    - Note: We distinguish between several language localizations, e.g. `de-AT` and `de-DE`. If no translation is available for the localization selected in the MOC, the system does not select an alternative localization, but displays the standard language (generally English).

Further information

- MDS-BAS_81: Multilingual texts in the MOC using language keys
- MDS-BAS_81: Activating the MES Development Suite
- MDS-BAS_81: Configuration settings and configuration levels

1.11 Logging and debugging

Further information

- CUT-MOC: Documentation of the training CUT-MOC: Information on the troubleshooting on the client and server.

1.12 Adding and customizing fields in a detail application

You can change fields in a detail application to improve the presentation of contents.

Requirements

- "MDS Development Suite" is enabled to configure applications.
- You have selected the correct scope to configure applications.
- The detail application that you want to change has been generated (by default, the selection panel is included in an application).
- The field you want to add must be available in the service, i.e. the field is an existing field or a customer-specific field that has been added to the service definition in the repository client.

Solution: Steps to follow

Add a field

1. Open the application you want to modify.
2. Right-click the layout of the detail application - "Add item".
3. Right-click the layout of the detail application - "Customize layout"
4. Select the newly created element (usually at the bottom of the detail application or the selection panel).
5. In the dialog "Customization", change to "Layout tree view".
6. Move the new field by drag and drop to the required position in the dialog "Customization" or directly in the application.
    - Change the width or height of a field by right-clicking the field in the list of the "Customization" dialog. Select "Size constraints" and "Free sizing".
    - Once you have adjusted the fields, set the "Size constraints" of all fields to "Lock size".
    - You can create an element at the bottom ("Create empty space item") to avoid that you unintentionally change the size of the last field.
7. Assign a "FieldName" in the dialog "Customization".
    - Use the service acronym to complete the configuration options automatically.
    - If necessary, you can define a context in form of a "ServiceName".
    - Or you must manually fill in all properties (detailed description in the documentation).
8. Save the application by clicking the button "Save".
9. If necessary, close and reopen the application to load the field configuration via service documentation.

Request field data from service

Note: If you proceed as follows, the name of the new field must match the name of the service field.

1. Entry in the data source

    10. Open the dialog "Configure data sources"
    11. Select the data source that should provide the requested data.
    12. Click "Configure".
    13. Click "Select columns".
    14. Select the column for which you have newly defined the field in the application.
2. Entry in the detail application
    10. Open the dialog "Configure detail application".
    11. Select the detail application that includes the newly defined field.
    12. Click "Configure".
    13. Click "Select columns".
    14. Select the column for which you have newly defined the field in the application.
3. Save the application by clicking the button "Save".

Edit an existing field

1. Open the application you want to modify.
2. Right-click the layout of the detail application - "Customize layout"
3. Select the field you want to modify.
4. Change field parameters in the dialog "Customization" (detailed description in the documentation).
5. Reposition the field in the application by drag and drop.
6. If required, edit the configuration of the data sources or the detail application (see above).
7. Save the application by clicking the button "Save".

Delete an existing field

1. Open the application you want to modify.
2. Right-click the field you want to delete and select "Delete item".
3. Save the application by clicking the button "Save".

Affected files

The changes described here affect the following files in the respective application directory <MOC>\local\conf\MOC\Apps\:

- The file Layout<ID of the detail application>.config if the detail application has been customized.

Restrictions

You can only configure fields of detail applications of type "layout" and of the selection panel. If you want to change other types of detail applications like charts or table views, please refer to the respective tutorials.

Possible problems

Note: The mere configuration of a new field is usually not sufficient. In addition, you must edit linked data sources or you must make available the new field in the detail application (see above).

Further information

- MDS-BAS_81: Customizing of main applications, section "Selection panel"
- MDS-BAS_81: Input and output fields (to assign field properties)
- Tutorial "Configuration of data sources"
- Tutorial "Configuration of selection fields"

1.13 Configuration of data sources

You can create or modify label texts for MOC applications to integrate customer-specific terminology or translations.

Requirements

- "MDS Development Suite" is enabled to configure applications.
- You have selected the correct scope to configure applications.

Solution: Steps to follow

1. Open the application you want to modify.
2. Open the dialog "Configure data sources" by clicking the button.
3. Click the required button, e.g. "Add".
    1. Enter "ID" of the data source as unique key.
    2. Select the "Data logic". It defines the requested web service.
    3. "Events" specify when the data source is requested, e.g. line break in the grid (`SelectionChanged` of another data source) or when clicking the button "Request data" (`DataRequested`).
    4. The "Source" defines possible selection parameters, e.g. "Parameter layout" to transfer data from the selection panel.
        1. For the input from the selection panel, "Parameter" specifies the field name in the selection panel, i.e. which user input is used.
        2. "Target" specifies the service field to which the user input is forwarded.
        3. "Default operator" defines the type of operator; if nothing is entered, the `EqualOperator` is used.
    5. Click the button "Select columns" to define the columns requested from the server.
    6. Confirm by clicking "OK".

Affected files

The changes described here affect the following files in the respective application directory `<MOC>\local\conf\MOC\Apps\`:

- The file `EventLinkCollection.config` to map events within the application.
- The file `DataControllerCollection.config` which includes the configuration of data sources and the requested columns.

Possible problems

- Problem: You cannot delete the data source.
    - Proceed as follows: Before trying to delete the data source, make sure that the data source is not in use.
    - Note: Data sources are not only used in detail applications, they can also be referenced by other data sources to provide parameters or in an event.

Further information

- MDS-BAS_81: Customizing of main applications, section "Data sources"

1.14 Configuration of detail applications

Modify detail applications of an application to implement customer-specific layouts or other contents.

Requirements

- "MDS Development Suite" is enabled to configure applications.
- You have selected the correct scope to configure applications.
- You have configured the data source for the detail application (see tutorial "Configuration of data sources").

Solution: Steps to follow

Create a new detail application

1. Open the application you want to modify.
2. Add a new detail application.
    3. Open the dialog "Configure detail application" by clicking the button.
    4. Click "Add".
        1. Define a unique ID.
        2. Define a caption according to the schema `lkU_xxx`.

3. Select the application category. The category specifies the design of the detail application. The most common designs are: layout, grid, chart (refer to the documentation for further information).
4. Select the configured data source.
5. In case of a layout application showing a detail view of a table: Enable the option "Show selected data records only".

5. Click "Select columns" to define the available fields of the data source for the application.
6. Create the new detail application by clicking "OK".

3. Position the new detail application within the application via docking (see tutorial "Customize application layout via docking")
4. Save the application by clicking the button "Save".

Edit existing detail applications

1. Open the application you want to modify.
2. Open the dialog "Configure detail application" by clicking the button.
3. Select the respective detail application.
4. Click "Configure" to edit the detail application.
    1. Edit the configuration (refer to the documentation for further information).
    2. Click "OK" to confirm your changes.
5. Save the application by clicking the button "Save".

Delete an existing detail application

1. Open the application you want to modify.
2. Undock the detail application by drag and drop.
3. Open the dialog "Configure detail application" by clicking the button.
4. Select the respective detail application.
5. Click "Remove" to delete the detail application.
6. Save the application by clicking the button "Save".

Affected files

The changes described here affect the following files in the respective application directory `<MOC>\local\conf\MOC\Apps\`:

- The file `DataControllerCollection.config` which includes the configuration of the application's data sources.
- The file `ApplicationPluginCollection.config` which includes the configuration of the application's detail application.
- The file `Layout<ID of the detail application>.config` as layout file of the respective detail application.

- If required, the file `DockManagerCollection.config` which saves the docking of the new detail application.

Further information

- MDS-BAS_81: Customizing of main applications, section "Detail applications"
- Tutorial "Customize application layout via docking"

1.15 Adding and customizing fields in the selection panel

You can change fields in the selection panel to improve the presentation of contents.

Requirements

- "MDS Development Suite" is enabled to configure applications.
- You have selected the correct scope to configure applications.
- The field you want to add must be available in the service, i.e. the field is an existing field or a customer-specific field that has been added to the service definition in the repository client.

Solution: Steps to follow

Add a field

1. Open the application you want to modify.
2. Right-click the selection panel - "Add item".
3. Right-click the selection panel - "Customize layout".
4. Select the newly created element (usually at the bottom of the detail application or the selection panel).
5. In the dialog "Customization", change to "Layout tree view".
6. Move the new field by drag and drop and assign the properties (for details, refer to the tutorial "Adding and customizing fields in a detail application").
7. Save the application by clicking the button "Save".
8. If necessary, close and reopen the application to load the field configuration via service documentation.

Use entered values as filter parameters

1. Entry in the data source
    1. Open the dialog "Configure data sources"
    2. Select the data source you want to query when clicking "Request data".
    3. Click "Configure".
    4. In the group "Parameter", the source "Parameter layout" is selected.
    5. In the grid, select the row with the values "Source" = `ParameterLayout` and "Parameter" = Name of the newly defined field.

6. In the column "Target" of this row: Enter the service parameter to which this parameter value is to be transferred.

2. Save the application by clicking the button "Save".

Edit an existing field

1. Open the application you want to modify.
2. Right-click the selection panel - "Customize layout".
3. Select the field you want to modify.
4. Change field parameters in the dialog "Customization" (detailed description in the documentation).
5. Reposition the field by drag and drop.
6. If required, edit the configuration of the data source (see above).
7. Save the application by clicking the button "Save".

Delete an existing field

1. Open the application you want to modify.
2. Right-click the field you want to delete and select "Delete item".
3. Save the application by clicking the button "Save".

Affected files

The changes described here affect the following files in the respective application directory `<MOC>\local\conf\MOC\Apps\`:

- The file `LayoutPanel.config` which is the configuration file of the selection panel.

Possible problems

Note: The mere configuration of a new field is usually not sufficient. In addition, you must edit linked data sources (see above).

You cannot use all parameters that are available in a service as filter parameters in the selection panel. In the service definition of the repository, set the value `isFilterParameter` to `true`.

Further information

- MDS-BAS_81: Customizing of main applications, section "Selection panel"
- MDS-BAS_81: Input and output fields
- Tutorial "Configuration of data sources"
- Tutorial: "Adding and customizing fields in a detail application"

1.16 Configuration of a dependent data source

Configure a data source that refers to another data source to request data (master-detail relationship).

Requirements

- "MDS Development Suite" is enabled to configure applications.
- You have selected the correct scope to configure applications.
- The master data source has already been created (see tutorial "Configuration of data sources").

Solution: Steps to follow

Example "Order overview": Selection of all operations (detail data source) of an order (master data source). The order number (order.id) is used for mapping. All operations of the selected order are then displayed in the detail table.

1. Open the application you want to modify.
2. Open the dialog "Configure data sources" by clicking the button.
3. Click the required button, e.g. "Add".
    1. Enter "ID" of the data source as unique key.
    2. Select the "Data logic". It defines the requested web service.
    3. The "Events" specify when the data source is requested. Here, you must select `<master data source>: SelectionChanged`.
    4. As "Source" it might be useful to select the master data source including the corresponding parameter mapping.
    5. Click "OK" to confirm data input.
4. Save the application by clicking the button "Save".

Affected files

The changes described here affect the following files in the respective application directory `<MOC>\local\conf\MOC\Apps\`:

- The file `EventLinkCollection.config` to map events within the application.
- The file `DataControllerCollection.config` which includes the configuration of data sources and the requested columns.

Further information

- MDS-BAS_81: Customizing of main applications, section "Data sources"
- Tutorial "Configuration of data sources"

1.17 Creating or editing the menu

You can create a custom menu to have quick access to important entries and integrate custom applications into the menu.

Requirements

- You have selected the correct scope.

Solution: Steps to follow

Create a custom menu

1. In the MOC, select "Extras" - "Menu editor".
2. Select the menu you want to edit in the drop-down list. Open this menu by clicking "Load".
3. We recommend not to overwrite the two standard menus, but to create a custom menu.
    1. Load the menu you want to edit.
    2. Define a name in the drop-down.
    3. Save
4. Edit the loaded menu (refer to the documentation for further information):
    - via buttons (add menu, delete entry),
    - edit text directly in the tree structure or
    - add new applications from the function list on the left by drag and drop.
    - Define parameters by clicking "Advanced settings".
5. Save the modified menu.
6. Select the menu via "Extras" - "Configuration" - "Select menu". The selected menu is filed in the user scope.

Add a custom application to the menu

1. Create a custom application (see tutorial "Generate applications")
2. Call the new application via transaction code `_capp id=<application name>` for test purposes.
3. Enter an authorization key if you want to avoid that any user can open the application.
    - Create a new entry in the file <MOC directory>\local\resources\data\authorizations\LocalApps.Authorization.xml.
    - AuthorizationId: Assign the ID of the application; AuthorizationKey: Assign an authorization key of max. 15 characters.
    - If the file or folder structure is not available, use the menu template of the training documentation to create it.
4. Enter a function authorization for the authorization key in order to assign authorizations to users.
    - Create a new entry in the file <MOC directory>\local\resources\data\authorizations\FunctionAuthorisationMappings\functionauthorisation.txt.
    - The syntax is: <function authorization>;<authorization key>
    - If the file or folder structure is not available, use the menu template of the training documentation to create it.
5. Enter the application in the list of available applications.

    - Create a new entry in the file <MOC directory>\local\resources\data\functions\functions.xml.
    - Use the ID of the application as LinkId and the name as Label. As Parameter, enter the id=<application ID>. Optionally, you can define a transaction code.
    - If the file or folder structure is not available, use the menu template of the training documentation to create it.
6. Optional: Test the defined transaction code (an error message pops up because of missing authorization).
7. Optional: Assign the required authorization to the current user for test purposes.
    1. On the MOC: "System administration" - "User admininstration" - "Function authorizations"
    2. Add a new entry to the current user and assign the function authorization from `functionauthorisation.xml`.
8. Restart the MOC.
9. Add the application to the menu, see "Creating a custom menu".

Affected files

Each menu is stored in a specific directory in `<MOC directory>\local\conf\MOC\Menues`. Each menu includes a main file that references the files of the individual groups of the menu. Each group again includes a subdirectory with one file per menu entry.

Possible problems

- Problem: You cannot select the new menu in "Extras" - "Configuration".
    - Proceed as follows: Make sure that you are in the correct scope (local or user).
- Problem: After having edited the XML files of the menu manually, the MOC does not start.
    - Proceed as follows: Make sure that the configuration files do not include inconsistencies, e.g. ensure that the indicated number of elements matches the actual number of existing entries.
    - Note: We generally recommend to use the menu editor to avoid corruption of the menu configuration.
- Problem: The menu editor does not show your custom application.
    - Proceed as follows: Make sure that you have added the correct entries to all three files (authorization key, mapping to function authorization and you have made the application available for the menu). It is of special importance that the authorization keys of `LocalApps.Authorization.xm` and `functionauthorisation.xml` are identical.
    - Tip: Assign a transaction code for test purposes and check, if you can call the application using this code (see above).
    - Note: You must restart the MOC to enable the changes in these files.

Further information

- MDS-BAS_81: Customizing files for menus
- Tutorial "Generating applications"

1.18 Customizing application layout via docking

You can position the detail applications in an MOC application and thus create a customer- or user-specific application layout.

Requirements

- "MES Development Suite" is enabled.
- You have selected the correct scope to configure applications.

Solution: Steps to follow

1. Move the detail application by drag and drop to the required position.
2. Save the changed application layout by clicking the toolbar button "Save".

Affected files

The changes described here affect the following files:
- <MOC directory>\local\conf\MOC\Apps\<application name>\DockManagerCollection.config

Possible problems

- Problem: The application layout is misconfigured and you want to return to the original state. The application is still open.
    - Proceed as follows: Close the application without clicking the "Save" button. All changes since the last time the application was saved are discarded.
    - Note: We recommend to make a backup copy of the application directory before editing the layout.

Further information

- MDS-BAS_81: Customization of application layout
- MDS-BAS_81: Activating the MES Development Suite
- MDS-BAS_81: Configuration settings and configuration levels

1.19 Customizing symbols and images

You can replace symbols and images with customer-specific icons, start screens and logos.

Requirements

- You have selected the correct scope.

Solution: Steps to follow

1. Adjust size of the new graphic (refer to the documentation for details).
2. Save the graphic with an appropriate name (case sensitive; refer to the documentation for specifications).
3. File in the directory `<MOC directory>\local\resources\images`.
4. (Re)start the MOC.

Affected files

When customizing symbols and images, the original files are not overwritten. Only the files in the directory `<MOC directory>\local\resources\images` are modified or added.

Possible problems

- Problem: The graphic has been copied to the correct storage location but it is not loaded.
  - Proceed as follows: Make sure that the file name is properly spelled (case sensitive, file type) and that the file is stored in the local or user scope.

Further information

- MDS-BAS_81: Customization of symbols
- MDS-BAS_81: Configuration settings and configuration levels

1.20 Adding and customizing buttons in the toolbar

You can add a new button to the toolbar or configure an existing button.

Requirements

- "MES Development Suite" is enabled.
- You have selected the correct scope to configure applications.

Solution: Steps to follow

1. Open the "Link editor" by calling the context menu in the toolbar and clicking "Configuration".
2. Add a new ApplicationCommandLink by clicking "New".
    1. Assign any ID and command.
    2. Select function. Possible values: see documentation.
    3. Enter the parameters matching the function.

4. Specify via the "Authorization key" if a button is enabled or grayed out.
5. The authorization key entered in the field "Visibility" controls if a button is displayed or not.
6. You must enter a language key in the field "Title". If the field "Tooltip" is empty, the value of "Title" is taken over when saving the changes.
7. "Category" and "Ribbon page" specify the position of the button within the toolbar.
8. "Disabled".
9. The button "Shortcut" opens a dialog to define a shortcut that runs the ApplicationCommandLink when the application is active.
10. If you check the option "Disabled", you can hide an existing button (of a higher scope). If you want to reactivate the button, you must do this in the configuration file.
11. You can configure an image from the MOC resources for the button layout. It depends on the size of the toolbar and the number of buttons to be shown, if the small or the large image is displayed. You cannot influence or change this.

3. Confirm the changes by clicking "OK".
4. You must save the application configuration. Restart the application to enable the configuration.

Affected files

The button configuration is stored within the application configuration in the file `ApplicationCommandLinkCollection.config`. Each button requires separate configuration.

Restrictions

You cannot customize buttons of the groups "Data", "Customizing", "Settings" and "Help" as described above.

Further information

- MDS-BAS_81: Toolbar
- MDS-BAS_81: Activating the MES Development Suite
- MDS-BAS_81: Configuration settings and configuration levels
- MDS-BAS_81: Authorization

1.21 Configuring a detail application of type "chart"

You can configure a detail application as chart to visualize data in a graphic manner.

Requirements

- "MDS Development Suite" is enabled to configure applications.
- You have selected the correct scope.
- You have configured the data source providing data for the chart.

Solution: Steps to follow

1. Open the application you want to modify.
2. Open the dialog "Configure detail application" by clicking the button.
3. Create a new detail application by clicking "Add".
    - Define an ID that is unique in the application.
    - Assign a caption as language key, e.g. `lkU_myChart`.
    - Select the application category `Chart`.
    - Select the data source that provides the necessary data to fill the chart.
4. Select the columns you want to integrate in the application by clicking "Select columns".
5. Position the chart in the application via docking.
6. Save the changed application by clicking the button "Save".
7. Before configuring the chart, request data once.
8. Open the chart wizard by double-clicking the chart.
    1. Select the chart type and confirm by clicking "Next".
    2. Select the "appearance" and confirm by clicking "Next".
    3. Define the "series" to be displayed. Define one series for each data source field you want to display. Confirm by clicking "Next".
    4. Configure "Data":
        - Define the "Value" of the field to be shown for this series, e.g. yield of an order.
        - Define the "Argument" for the grouping of values, e.g. grouping of yield per person.
    5. [Optional] Define further layout properties of the chart.
    6. Confirm by clicking "Finish".
9. Save the application by clicking the button "Save".

Refer to the documentation for details on the configuration of special chart types.

Affected files

The changes described here affect the following files in the respective application directory `<MOC>\local\conf\MOC\Apps\`:

- The file `ApplicationPluginCollection.config` which includes the configuration of the application's detail application.
- The file `DockManagerCollection.config` which saves the docking of the new detail application of type "chart".
- The file `Chart<ID of the detail application>.config`: the configuration of the detail application.
- The file `DataControllerCollection.config` which includes the configuration of the application's data sources if the service requests further columns.

Further information

- MDS-BAS_81: Components for detail applications, section "Charts"
- Tutorial "Configuration of data sources"
- Tutorial: "Configuration of detail applications"

1.22 Configuring a detail application of type "table"

You can configure a detail application as grid to visualize data in tabular form.

Requirements

- "MDS Development Suite" is enabled to configure applications.
- You have selected the correct scope.
- You have configured the data source providing data for the grid.

Solution: Steps to follow

Adding a detail application of type "grid"

1. Open the application you want to modify.
2. Open the dialog "Configure detail application" by clicking the button.
3. Create a new detail application by clicking "Add".
    - Define an ID that is unique in the application.
    - Assign a caption as language key, e.g. `lkU_myGrid`
    - Select the application category `Tabellenansicht`
    - Select the data source that provides the necessary data to fill the table.
4. Select the columns you want to integrate in the application by clicking "Select columns".
5. Position the grid in the application via docking.
6. Save the changed application by clicking the button "Save".

Refer to the documentation for details on the configuration of the table grid.

Adding a column to the grid

If a new field is available in the service and you want to display this field in the detail application of type "table", proceed as follows:

1. Open the application you want to modify.
2. Request the new field of the service.
    1. Open the dialog "Configure data sources", select the data source defined for the grid.
    2. Click "Configure" to edit the data source.
    3. Open the dialog "Select columns" by clicking the button.
    4. Select the new field and confirm the changes.

3. Make the field available in the detail application.
    1. Open the dialog "Configure detail application", select the grid.
    2. Click "Configure" to edit the grid.
    3. Open the dialog "Select columns" by clicking the button.
    4. Selct the new field and confirm the changes.
4. Add the column in the grid.
    1. Right-click the grid column, "Add column".
    2. Define the new column.
        - Name: define unique name.
        - Field name: Field name of the service, e.g. `order.id`
        - Caption: The header in the grid (defined by a language key).
        - Category: to group the columns.
5. Save the application by clicking the button "Save".

Affected files

The changes described here affect the following files in the respective application directory `<MOC>\local\conf\MOC\Apps\`:

- The file `ApplicationPluginCollection.config` which includes the configuration of the application's detail application.
- The file `DockManagerCollection.config` which saves the docking of the new detail application of type "table".
- The file `Grid<ID of the detail application>.config`: the configuration of the detail application which is created when you add a detail application of type "table".
- The file `DataControllerCollection.config`: the configuration of the application's data sources if further columns are added or requested by the service.

Possible problems

- Problem: A wrong language key has been assigned to the added column. The language key must be changed.
    - Proceed as follows: Right-click the grid column - "Grid properties". Select the proper field in the drop-down list. Change the property "LanguageKey" to the required value. Save the application.
    - Note: You must restart the application to enable the changes.

Further information

- MDS-BAS_81: Components for detail applications, section "Table view (grid)"
- Tutorial "Configuration of data sources"

- Tutorial: "Configuration of detail applications"

1.23 Configuring a detail application of type "pivot"

You can configure a detail application or type "pivot" to visualize data in a pivot table.

Requirements

- "MDS Development Suite" is enabled to configure applications.
- You have selected the correct scope.
- You have configured the data source providing data for the pivot table.

Solution: Steps to follow

6. Open the application you want to modify.
7. Open the dialog "Configure detail application" by clicking the button.
8. Create a new detail application by clicking "Add".
    - Define an ID that is unique in the application.
    - Assign a caption as language key, e.g. `lkU_myChart`.
    - Select the application category `PivotTabelle`
    - Select the data source that provides the necessary data to fill the pivot table.
9. Click "Select columns" to select the columns you want to transfer into the application, i.e. for the fields that are available in the pivot table.
10. Position the pivot table in the application via docking.
11. Save the changed application by clicking the button "Save".
12. Configure the pivot table.
    - Right-click "Drag filter fields here" - "Show all fields" or "Show field list". Drag and drop the fields required for the pivot table.
    - Assign the fields to column fields and row fields to group the data by these fields.
    - Assign the fields to data fields to display the data in the pivot table.
    - Configure the chart added below by double-clicking the chart (see tutorial "Configure a detail application of type "chart").
13. Save the changed application by clicking the button "Save".

Refer to the documentation for details on the configuration possibilities.

Affected files

The changes described here affect the following files in the respective application directory `<MOC>\local\conf\MOC\Apps\`:

- The file `ApplicationPluginCollection.config` which includes the configuration of the application's detail application.
- If required, the file `DockManagerCollection.config` which saves the docking of the new detail application of type "pivot".
- The file `Pivot<ID of the detail application>.config`: the configuration of the detail application.
- The file `DataControllerCollection.config` which includes the configuration of the application's data sources if the service requests further columns.

Further information

- MDS-BAS_81: Components for detail applications, section "Pivot"
- Tutorial "Configuration of data sources"
- Tutorial: "Configuration of detail applications"
- Tutorial: "Configuring a detail application of type 'chart'"

1.24 Protecting a field via authorization

You can protect a new field of an application and a service so that a missing server component cannot lead to an error in the client.

Requirements

- The field `a_acronym` has been defined in the application `A`
- The field `a_acronym` is available in the requested service of the domain `a_dom`.

Solution: Steps to follow

1. Use an authorization key to protect the field a_acronym.
    1. Create a new key in the repository client in the dialog "Authorization".
    2. Domain = `a_dom`
    3. Authorization Id = `a_acronym` (the field you want to protect)
    4. Authorization type = Acronym
    5. Authorization key: select an individual key `a_authkey`
    6. Export the data into the respective domain + runtime structure.
2. Define the mapping: authorization key - FeatureSet.
    1. In the domain MOC authorization (in products), edit the file %MOC authorization%bin\resources\data\authorizations\FeatureSetMappings\featureset.txt
    2. Add the entry a_featureSet;a_authkey. Here: a_featureSet specifies the new FeatureSet that you want to define.
3. Create the FeatureSet.

1. Check out the corresponding server domain.
2. Create a new FeatureSet mapping in the directory `%Service%\ConfigManager a_featureSetMappingFile.xml`
3. Define contents according to the documentation, e.g. check in the FeatureSet `a_featureSet` if a database patch is available. The field `a_acronym` that you want to protect is only activated, if the patch assigned in the FeatrueSet has actually been executed.
4. Edit the FeatureSet Excel file as list of all available FeatureSet mappings.

4. Deployment of the FeatureSet configuration on the server to `%InstallDir%\JDIR\MOC\1\configManager\standard`
5. Restart the Tomcat.

The client field is protected by the defined authorization key and only visible, if this key is enabled. The authorization key is enabled if the corresponding FeatureSet mapping is found and enabled. This is only the case if the customization has been imported into the server and the feature set is thus available.

Further information

- Documentation MDS-BAS